

Multi-Cohort Simulation Study: Shared vs. Study-Specific Factors

Andrew Walther

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1. Purpose and Background

Prior synthetic validation of the Supervised Bayesian Matrix Factorization (SSBMF) model was carried out only in a **single-cohort** setting. The real application pools several PDAC cohorts that differ both by **platform** (RNA-seq, microarray, proteomics) and by **biology**. In that setting the latent factors fall into two kinds:

- **Shared factors** — gene-expression programs present across *all* studies. These are the cross-cohort biological signal we want to detect and, where they are prognostic, to use for survival prediction.
- **Study-specific factors** — programs private to a *single* study. These absorb platform/batch effects and cohort-private biology that should *not* be mistaken for shared, generalizable signal.

This proposal specifies a simulation that generates data with a **known** shared / study-specific factor structure, so we can test whether the current model **natively** recovers that distinction — the same goal the previously discussed cohort dummy-indicator approach was meant to serve. If the base model already separates shared from study-specific factors, the dummy indicator columns may be unnecessary.

The design incorporates the plan from the last meeting (5/29), with two key elements:

1. **EBMF-grounded ground truth.** Realistic gene programs are taken from an Empirical Bayes Matrix Factorization (`flashier`) fit to the merged real training cohorts. EBMF is also our **unsupervised benchmark** (Section 5).
2. **Shared factors carry survival signal; study-specific factors do not** ($\beta \neq 0$ on shared, $\beta = 0$ on specific). Survival is generated from the shared component only.

2. Notation

Symbol	Meaning	Shape
C	number of cohorts (studies)	scalar
n_c	patients in cohort c ; $n = \sum_c n_c$	scalar
p	number of genes	scalar
K	total number of latent factors	scalar
$\mathcal{S}$	index set of shared factors	$ \mathcal{S} $
$\mathcal{P}_c$	index set of factors specific to cohort c	$ \mathcal{P}_c $
L	patient loadings	$n \times K$
F	gene programs (factors)	$p \times K$
β	log-hazard coefficients	K
Y	observed genomics matrix	$n \times p$
$a_{\text{shared}}, a_{\text{specific}}$	amplitude (signal strength) of shared / specific factors	scalar

Rows of Y (and L) are **stacked by cohort**: rows $1..n_1$ are cohort 1, the next n_2 rows are cohort 2, and so on. Factors partition disjointly: $K = |\mathcal{S}| + \sum_c |\mathcal{P}_c|$. The remaining noise and survival parameters ($\tau_j, \alpha, \lambda_0, U_i$) are defined inline where they first appear in Section 3.

3. Data-Generating Process

We *specify* the latent quantities L , F , and β — building in exactly which factors are shared and which are cohort-specific, and which carry survival signal — then *generate* the observed data (Y , time, status) from them. The model is given only the observed data and tasked with recovering the latent structure; because we know the ground truth, every recovery error is directly measurable.

Specified (ground truth): F (gene programs), L (patient loadings with the shared/specific block structure), β (which factors are prognostic). **Generated (model input):** Y (expression matrix), and the survival outcomes time and status. **Model’s task:** recover $\hat{F}$, $\hat{L}$, and $\hat{\beta}$ from $(Y, \text{time}, \text{status})$ alone.

The components are described below in the order they are constructed.

3.1 Gene Programs F ($p \times K$) — what each factor “looks like” in gene space

F is specified first because it defines what each latent factor *means* biologically: column k is a p -vector of gene weights describing the k -th gene expression program.

Construction. We use real PDAC gene programs as ground truth rather than arbitrary synthetic ones. Concretely, we fit an Empirical Bayes Matrix Factorization (EBMF; flashier, var_type = 2, greedy + backfit) to the merged real PDAC training cohorts and extract the resulting gene-loading matrix — $p \times K_{\text{EBMF}}$ columns, each a sparse, data-grounded gene program. These columns are then used as the K columns of F (recycled by index if $K > K_{\text{EBMF}}$).

Two adjustments are made before use:

1. **Unit normalization.** Each column is divided by its L_2 norm so that $\|F_{\cdot k}\| = 1$. This decouples program *shape* from program *amplitude*, letting a separate scalar a_k control how strongly factor k appears in Y .
2. **Non-negativity.** The SSBMF model places a non-negative (point_exponential) prior on F — the same NMF-style assumption used at initialization. The EBMF programs are signed; taking their absolute value gives non-negative programs consistent with the model’s representational assumption. The EBMF benchmark arm is run non-negative for the same reason.

Each column is then multiplied by an amplitude scalar: a_{shared} for columns assigned to shared factors and a_{specific} for cohort-specific ones. Because $\|F_{\cdot k}\| = 1$, the variance contributed by factor k to Y scales as a_k^2 , making the amplitude the direct signal-strength dial.

A synthetic fallback is used when real data are unavailable: each column is a random sparse vector (5% of genes active, equal weights), unit-normalized and sign-adjusted the same way.

3.2 Patient Loadings L ($n \times K$) — who expresses each factor, and how much

L encodes how strongly each patient expresses each gene program. The entry L_{ik} is the *loading* of patient i on factor k : large values mean patient i ’s expression profile is strongly shaped by program k .

Distribution. Every entry is drawn independently from an Exponential distribution with rate 1:

$$L_{ik} \sim \text{Exponential}(\text{rate} = 1), \quad \mathbb{E}[L_{ik}] = 1, \quad \mathbb{E}[L_{ik}^2] = 2.$$

The Exponential is non-negative and right-skewed: most patients have a moderate loading near 1, a minority have high loadings, and none are negative. This matches the point_exponential prior the model places on L during inference — the ground truth and the prior are on the same scale and in the same family.

Shared vs. specific structure. After drawing all entries, specificity is imposed by zeroing out patients who do not belong to the owning cohort:

- **Shared factor** $k \in \mathcal{S}$: L_{ik} is kept for *every* patient, regardless of cohort. All patients can express a shared program.
- **Cohort- c -specific factor** $k \in \mathcal{P}_c$: L_{ik} is set to **zero** for every patient i not in cohort c . Only patients from cohort c can express this program; it is invisible in all other cohorts.

The resulting L is block-sparse: shared columns are dense (all patients), specific columns are zero everywhere except one cohort block. This is the operational definition of “study-specific” built into the simulation.

3.3 Survival coefficients β (K -vector) — which factors are prognostic

β is **fixed by design** — it is not drawn from a distribution. It encodes which factors the model should identify as prognostic:

- **Shared factors** $k \in \mathcal{S}$: assigned nonzero coefficients, cycling the pattern $(1.5, -1.2, 0.8, -0.5)$ to length $|\mathcal{S}|$. For the primary hybrid scenario ($|\mathcal{S}| = 2$): $\beta = (1.5, -1.2)$, so the first shared factor is positively prognostic (high loading $\rightarrow$ worse outcome), the second is protective.
- **Cohort-specific factors** $k \in \mathcal{P}_c$: $\beta_k = 0$. These factors drive expression patterns within a cohort but carry no survival information. The correct model behaviour is to assign $\hat{\beta}_k \approx 0$ to these factors.

The model receives no information about which β_k are zero — it must infer this from the data. Recovering nonzero β on shared factors (true positives) while keeping $\hat{\beta} \approx 0$ on specific factors (avoiding false positives) is the central evaluation criterion.

3.4 Observed genomics matrix Y ($n \times p$)

Given L, F , we generate what the model will actually observe:

$$Y = LF^\top + E, \quad E_{ij} \sim \mathcal{N}(0, 1/\tau_j), \quad \tau_j \sim \text{Gamma}(\text{shape} = 2, \text{rate} = 2).$$

Each gene j has its own noise precision τ_j drawn from a Gamma prior ($\mathbb{E}[\tau_j] = 1$, so typical noise SD ≈ 1). This gives heteroscedastic noise across genes, matching real expression data. Y is column-centered before being passed to the model (standard preprocessing).

At this point the simulation has produced the expression matrix Y — equivalent to what we would observe from an RNA-seq or array experiment — with the latent factor structure (L, F) baked in.

3.5 Observed survival outcomes — time and status

The survival outcomes are generated from the **shared factors only**, via a Weibull proportional-hazards model. Specific factors are deliberately excluded from the survival model ($\beta_k = 0$), so any prognostic signal the model finds on specific factors is a false positive.

Linear predictor (shared factors only):

$$\eta_i = \sum_{k \in \mathcal{S}} L_{ik} \beta_k.$$

Event time via Weibull-PH inverse CDF:

$$T_i = \left(\frac{-\log U_i}{\lambda_0 \cdot \exp(\eta_i)} \right)^{1/\alpha}, \quad U_i \sim \text{Uniform}(0, 1),$$

where $\alpha = 1.5$ (Weibull shape) and $\lambda_0 = 0.01$ (baseline scale). This is the standard inversion method: U_i is a uniform random draw, and the formula inverts the Weibull-PH survival function $S(t | \eta_i) = \exp(-\lambda_0 t^\alpha e^{\eta_i})$ to produce a random event time that respects the proportional-hazards structure.

Censoring is applied by drawing a censoring time $C_i = s \cdot B_i$ where $B_i \sim \text{Exponential}(1)$ and s is a scale chosen by bisection to achieve approximately 30% censoring:

$$\text{time}_i = \min(T_i, C_i), \quad \text{status}_i = \mathbf{1}\{T_i \leq C_i\}.$$

What the model receives. The model is given only $(Y, \text{time}, \text{status})$ — the same inputs it would receive from a real study. It does not see L, F , or β . Its task is to jointly factorize Y and fit the survival model to recover $\hat{L}, \hat{F}$, and $\hat{\beta}$.

Anchoring. Because $\beta \neq 0$ only on shared factors, survival supervision pulls those factors toward the shared component — the model has an outcome-level incentive to represent them accurately. This is the intended mechanism and the key advantage of supervised over unsupervised factorization.

4. The Three Scenarios

The scenarios differ only in how the K factors are partitioned into shared vs. specific. For the primary setting $C = 2$, $n_c = 150$, $p = 1000$, $K = 6$:

Scenario	$ \mathcal{S} $ shared ($\beta \neq 0$)	specific per cohort ($\beta = 0$)	Interpretation
All shared	6	0, 0	Studies effectively one cohort
Hybrid	2	2, 2	Both shared and private structure
Nothing shared	0	3, 3	No cross-cohort structure

Nothing-shared is a deliberate false-positive control. With $\mathcal{S} = \emptyset$, $\eta \equiv 0$ and survival carries no factor signal. The correct result is C-index ≈ 0.5 and all $|\hat{\beta}|$ below threshold — the model should *not* fabricate a shared prognostic factor. An optional knob (`specific_prognostic`, default off) can give each cohort a within-cohort prognostic factor if we later want survival signal in this regime.

Worked example — Hybrid, $C = 2$, $n_c = 150$, $p = 1000$, $K = 6$

- **Dimensions:** $n = 300$; Y is 300×1000 , L is 300×6 , F is 1000×6 , β is length 6. Rows 1–150 = cohort 1; 151–300 = cohort 2.
- **Factor labels:** [shared, shared, specific_1, specific_1, specific_2, specific_2].
- **Loadings L :** columns 1–2 dense (all 300 patients). Columns 3–4 zero for rows 151–300 (private to cohort 1). Columns 5–6 zero for rows 1–150 (private to cohort 2). L shows a visible block-sparse pattern.
- **Coefficients:** $\beta = [1.5, -1.2, 0, 0, 0, 0]$ — only the two shared factors drive survival.
- **What a correct model should find:** factors 1–2 with loadings across *both* cohorts *and* nonzero $\hat{\beta}$; factors 3–6 cohort-private with $\hat{\beta} \approx 0$. The unsupervised EBMF benchmark should recover the same six gene programs but cannot say which two are prognostic.

Secondary sensitivity: a 3-cohort hybrid (2 shared + 1 private per cohort), and a sweep of the shared-to-specific amplitude ratio ($a_{\text{shared}}/a_{\text{specific}}$) within the hybrid regime to stress the case where study-specific signal dominates.

5. Model Arms

Each simulated dataset (Y , time, status, cohort_id) is fit by **five arms** — four supervised (LB/YFB $\times$ $\pm$ cohort indicator) plus the unsupervised EBMF benchmark:

Arm	Model	Cohort indicator
YFB_base	YFB ($\eta = (YF)\beta$) — recommended	none
YFB_cohort	YFB	yes
LB_base	LB ($\eta = L\beta$)	none
LB_cohort	LB	yes
EBMF	unsupervised flashier (survival-blind)	—

All supervised arms use a non-negative (`point_exponential`) prior on L and F and a normal prior on β . The cohort indicator, when present, absorbs per-cohort gene-mean differences without entering the survival predictor.

The native-identification test. The headline analysis is the **base** arms (no cohort indicator): can the fitted factors separate into shared vs. study-specific on their own? The $\pm$ indicator comparison then quantifies whether the indicator adds anything.

Why EBMF. EBMF is fit to the same Y but is survival-blind, so it has no β . It is the unsupervised gold standard for the factorization itself, answering two questions:

1. Does the supervised model recover the shared/study-specific structure *as well as* EBMF (i.e., supervision does not cost structural fidelity)?

2. Does supervision *additionally* yield correct β — flagging which factors are prognostic — which EBMF fundamentally cannot?

This is the value-add argument: same structural recovery, plus correct survival attribution.

The EBMF here (fit to each *simulated* Y , scored as a competing method) is distinct from the EBMF in Section 3.2 (fit to *real* data to build ground-truth programs). They do not share a fit.

6. Evaluation Metrics

For each (scenario $\times$ model arm $\times$ seed) we record:

- **Shared-factor recovery** — mean best $|\text{cor}|$ between estimated and true gene programs (F columns, gene space, comparable across LB/YFB/EBMF), over the *true shared* factors.
- **Specific-factor recovery** — same, over the *true study-specific* factors.
- **Specificity-classification accuracy** — label each *estimated* factor shared/specific from its per-cohort loading energy (a factor is called specific if more than 85% of its squared-loading energy concentrates in a single cohort), then compare to the truth. This is the direct test of “does the model natively identify the distinction?”
- **Survival-effect recovery** — $|\hat{\beta}|$ on factors matched to true shared (expect above a small threshold) vs. true specific (expect ≈ 0). Reported as true-positive and **false-positive prognostic rates** (the latter is the anchoring / nothing-shared check).
- **C-index** — orientation-free held-out concordance, $\max(C, 1 - C)$.

Comparisons reported: (i) YFB vs. LB; (ii) base vs. +cohort-dummy (Δ in each metric — “are dummies necessary?”); (iii) supervised vs. unsupervised EBMF.

7. Reference Data-Generating Code

The following is the complete, self-contained DGP (the project implementation wraps this same logic with configuration and seeds). A reader can run it directly given the small `.calibrate` helper.

```
# Bisection calibration of censoring scale to a target censoring fraction.
.calibrate <- function(ev, base, target = 0.30, n_iter = 40) {
  lo <- 1e-3; hi <- max(ev) * 10
  for (i in seq_len(n_iter)) {
    mid <- sqrt(lo * hi)
    if (mean(base * mid < ev) > target) lo <- mid else hi <- mid
  }
  hi
}

generate_multicohort_data <- function(
  C          = 2L,          # number of cohorts
  n_per      = rep(150L, C), # patients per cohort (rows stacked by cohort)
  p          = 1000L,       # genes
  K_shared   = 2L,          # # shared factors      (beta != 0)
  K_specific = rep(2L, C),  # # specific factors per cohort (beta = 0)
  F_templates = NULL,       # p x K_ebmf EBMF programs; NULL -> synthetic sparse
  a_shared   = 1.0,         # amplitude of shared F columns
  a_specific = 1.0,         # amplitude of specific F columns
  active_rate = 0.05,       # synthetic-fallback sparsity
  beta_shared = NULL,       # length K_shared; NULL -> recycle c(1.5,-1.2,0.8,-0.5)
  offset_sd   = 0.0,        # explicit platform offset SD (0 = off)
  shape      = 1.5, scale0 = 0.01, # Weibull-PH baseline (shape = alpha, scale0 = lambda_0)
  target_cens = 0.30,       # target censoring fraction
  seed       = 1L) {
```

```

set.seed(seed)
n <- sum(n_per)
K <- K_shared + sum(K_specific)
cohort_id <- factor(rep(seq_len(C), times = n_per))

# label + owner per column (owner 0 = shared)
labels <- c(rep("shared", K_shared),
            unlist(lapply(seq_len(C), function(cc) rep(paste0("specific_", cc), K_specific[cc]))))
owner <- c(rep(0L, K_shared),
           unlist(lapply(seq_len(C), function(cc) rep(cc, K_specific[cc]))))

# L (n x K): Exponential(1), block-zeroed outside owner cohort
L <- matrix(rexp(n * K, rate = 1), n, K)
for (k in seq_len(K)) if (owner[k] != 0L) L[cohort_id != owner[k], k] <- 0

# F (p x K): EBMF templates (unit-norm) or synthetic sparse (unit-norm)
Fm <- matrix(0, p, K)
if (!is.null(F_templates)) {
  idx <- ((seq_len(K) - 1L) %>% ncol(F_templates)) + 1L
  Fm <- F_templates[, idx, drop = FALSE]
  Fm <- sweep(Fm, 2, sqrt(colSums(Fm^2)) + 1e-10, "/")
} else {
  for (k in seq_len(K)) {
    active <- sample.int(p, max(1L, round(active_rate * p)))
    Fm[active, k] <- 1 / sqrt(length(active))
  }
}
amp <- ifelse(labels == "shared", a_shared, a_specific) # signal-strength knob
Fm <- sweep(Fm, 2, amp, "*")

# per-gene Gaussian noise: tau_j ~ Gamma(2,2)
tau <- rgamma(p, shape = 2, rate = 2)
E <- sweep(matrix(rnorm(n * p), n, p), 2, sqrt(tau), "/")

# optional explicit platform offset (off by default)
offset <- matrix(0, n, p)
if (offset_sd > 0) for (cc in seq_len(C)) {
  ov <- rnorm(p, 0, offset_sd)
  offset[cohort_id == cc, ] <- matrix(ov, sum(cohort_id == cc), p, byrow = TRUE)
}

# assemble + column-center
Y <- L %*% t(Fm) + offset + E
Y <- sweep(Y, 2, colMeans(Y), "-")

# survival from SHARED factors only
if (is.null(beta_shared))
  beta_shared <- if (K_shared > 0) rep_len(c(1.5, -1.2, 0.8, -0.5), K_shared) else numeric(0)
beta <- numeric(K); beta[labels == "shared"] <- beta_shared
eta <- as.vector(L %*% beta) # == 0 in the "nothing shared" scenario

event <- (-log(runif(n)) / (scale0 * exp(eta)))^(1 / shape)
cb <- rexp(n, rate = 1)
cs <- .calibrate(event, cb, target = target_cens)
ctime <- cb * cs
time <- pmin(event, ctime)
status <- as.integer(event <= ctime)

```

```
list(Y = Y, time = time, status = status, cohort_id = cohort_id,
     L_true = L, F_true = Fm, beta_true = beta,
     factor_labels = labels, factor_owner = owner)
}
```

8. Initial Results

The design above has been implemented and run end-to-end (3 scenarios $\times$ 5 arms $\times$ 5 seeds). This section gives a high-level overview; full tables, supplementary figures, verification, and the signal-ratio sweep appear in Part II (the remaining pages of this document). **These are preliminary and may be revised after this proposal is discussed.**

Scenario	Arm	Rec. shared	Rec. specific	Spec. acc.	C-index	β FP
all_shared	YFB_base	0.92	—	1.00	0.87	—
all_shared	YFB_cohort	0.94	—	0.90	0.86	—
all_shared	LB_base	0.53	—	1.00	0.74	—
all_shared	EBMF→Cox	0.82	—	0.90	0.86	—
hybrid	YFB_base	0.90	0.84	0.97	0.81	0.50
hybrid	YFB_cohort	0.92	0.92	1.00	0.81	0.20
hybrid	LB_base	0.72	0.67	0.83	0.76	0.20
hybrid	EBMF→Cox	0.68	0.78	0.80	0.72	—
nothing_shared	YFB_base	—	0.89	1.00	0.55	0.03
nothing_shared	YFB_cohort	—	0.91	1.00	0.54	0.03
nothing_shared	LB_base	—	0.78	0.87	0.54	0.07
nothing_shared	EBMF→Cox	—	0.70	0.80	0.55	—

Mean over 5 seeds; LB_cohort omitted for brevity. **Column definitions:** *Rec. shared / Rec. specific* — gene-program recovery, the mean best |correlation| between estimated and true gene programs over the true shared / study-specific factors (1 = perfect). *Spec. acc.* — specificity-classification accuracy, the fraction of estimated factors correctly labeled shared vs. study-specific from their per-cohort loading energy. *C-index* — held-out (external) concordance on the 25% test split (0.5 = chance). *β FP* — false-positive prognostic rate: fraction of true study-specific factors ($\beta = 0$) wrongly given a survival effect (lower is better). *EBMF→Cox* is the two-stage baseline: an unsupervised (survival-blind) EBMF factorization followed by a Cox model fit on the EBMF factor scores. It has no β false-positive rate (its factors are not chosen for prognostic relevance); its C-index quantifies how well the “factorize first, predict second” pipeline generalizes, for direct comparison against the joint supervised model. In the hybrid scenario the joint model leads by ≈ 0.09 (0.81 vs. 0.72); in all_shared the two-stage baseline nearly matches (every factor is prognostic), and in nothing_shared both sit at chance.

The headline findings:

- The model natively identifies the shared / study-specific distinction.** In the hybrid scenario, YFB classifies factors as shared vs. specific with accuracy 0.97 (base) to 1.00 (+cohort indicator), purely from per-cohort loading energy — no indicator columns required. This directly answers the central question.
- It does not fabricate prognostic signal when none exists.** In nothing_shared ($\eta \equiv 0$ by construction), every arm sits at C-index ≈ 0.54 and the β false-positive rate is 0.03–0.07 — the model does not invent a shared prognostic factor out of purely study-specific structure.
- The joint model beats the two-stage EBMF→Cox baseline where it matters most.** In the hybrid scenario — only some factors prognostic, the realistic case — the joint YFB reaches C-index 0.81 vs. 0.72 for “factorize first, fit Cox second” (≈ 0.09 gain). Because EBMF chooses its factors without seeing the outcome, it cannot preferentially sharpen the prognostic ones; jointly modelling genomics and survival can. In all_shared (every factor prognostic) the two-stage baseline nearly matches (0.86 vs. 0.87), and in nothing_shared both sit at chance — exactly where the gap should and should not appear. This is the core value-add of supervision.
- Supervision does not cost structural fidelity.** YFB recovers shared programs at |cor| ≈ 0.90 –0.94 and specific programs at 0.84–0.92, matching or exceeding the non-negative EBMF benchmark (0.68–0.82).
- The cohort indicator helps under genuine heterogeneity, and is redundant otherwise.** In hybrid, adding it halves the β false-positive rate (0.50 $\rightarrow$ 0.20) and lifts specificity accuracy to 1.00; in all_shared it adds nothing. This is the quantitative answer to “are the dummy indicators necessary?”

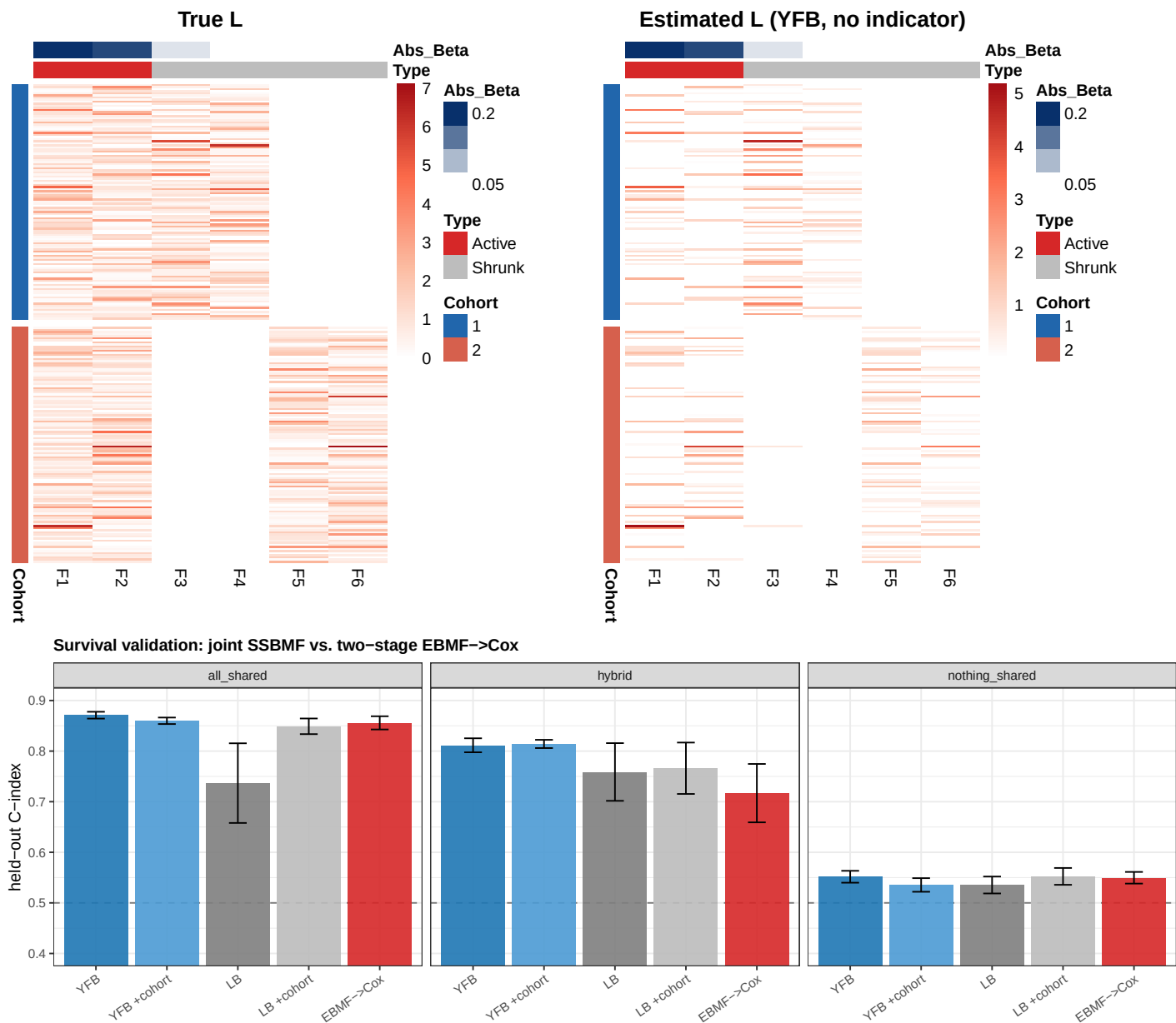


Figure 1: TOP — Patient loading heatmaps for the hybrid scenario (seed 1, training split). Rows = patients grouped by cohort (left sidebar: blue = cohort 1, salmon = cohort 2); columns = factors F1-F6. Top annotation: true factor type (Active = shared/prognostic; Shrunk = study-specific/non-prognostic) and estimated $|\hat{\beta}|$ from YFB. LEFT: true L (ground-truth block-sparse structure); RIGHT: estimated L from YFB recovering the same structure with no indicator columns. BOTTOM — Held-out C-index by scenario and arm (mean $\pm$ SE over 5 seeds; dashed line = chance 0.5). The joint supervised model (YFB) and the two-stage EBMF $\rightarrow$ Cox baseline differ most in the hybrid scenario: when only some factors are prognostic, jointly modelling genomics and survival recovers the prognostic structure that the survival-blind EBMF factorization cannot preferentially sharpen. In all_shared (every factor prognostic) the two-stage baseline nearly matches; in nothing_shared both sit at chance, as designed.

6. **YFB is more reliable than LB**, which collapses to near-zero recovery on some seeds — consistent with YFB being the recommended model.
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9. Open Questions for Discussion

1. **Specificity definition.** We encode study-specific factors via block-zero loadings (a factor is active in exactly one cohort). Is this the structure you intend, or should specific factors also have *non-overlapping gene programs* per cohort?
2. **Nothing-shared survival.** Default leaves survival null in the nothing-shared scenario (false-positive control). Do you prefer instead a within-cohort prognostic specific factor so each cohort has its own survival signal?
3. **Cohort count.** Primary is $C = 2$ (matches the real training data and the fixed-effects rationale); a $C = 3$ hybrid is included as sensitivity. Is 2 sufficient for the methods claim, or should 3 be primary?
4. **Signal ratio.** Default $a_{\text{shared}} = a_{\text{specific}}$. Should the hybrid scenario stress a regime where study-specific variance *dominates* shared variance (the harder, more realistic case)?
5. **Cohort indicator: design decision and remaining false positives.** In the hybrid scenario, adding the cohort dummy indicator halves the β false-positive rate ($0.50 \rightarrow 0.20$), but does not eliminate it — 20% of study-specific factors still receive a spurious prognostic coefficient even with the indicator present.

The mechanism is confounding: study-specific factors are active in exactly one cohort, and that cohort's patients have different survival trajectories (driven by the shared prognostic factors with different mean loadings across cohorts). Without a cohort label, the model absorbs this cohort-level survival difference by assigning $\hat{\beta}$ to the specific factors. The indicator breaks this confounding, but residual within-cohort correlation (finite sample, $n_c = 150$) accounts for the remaining 0.20.

Two open sub-questions:

- (a) *Should the cohort indicator be part of the default recommended configuration?* The current D4 real-data benchmark achieves $C = 0.636$ **without** the indicator, suggesting confounding is tolerable in practice. But this simulation targets the equal-signal hybrid ($a_{\text{shared}} = a_{\text{specific}}$); whether the PDAC cohorts sit in a harder regime (Q4 above) where the FP problem worsens is unknown.
- (b) *What additional approaches could reduce the residual false positives?* Three candidates:
 - **Signal-ratio sweep** (planned): stress the specific-dominant regime ($a_{\text{specific}} \gg a_{\text{shared}}$) to quantify how badly the FP rate grows and whether the indicator remains sufficient.
 - **Tighter β prior**: a tighter normal prior on β would shrink small spurious coefficients toward zero without the full collapse that the `point_normal` prior causes. This has not been tested in this setting.
 - **Cohort-stratified Cox baseline**: replace the shared baseline hazard with cohort-stratified strata, natively absorbing cohort-level survival differences without indicator columns. This is a more substantial model change.

The signal-ratio sweep (Q4/Q5b) has now been completed (see Part II, Section R6 below). Key findings: (i) the FP rate is bounded at 0.20–0.50 across all variance ratios — it does not grow monotonically as specific factors dominate; (ii) the indicator's advantage is regime-dependent and narrows at high variance ratios; (iii) structural recovery and C-index are completely stable across all ratios. The recommendation: include the cohort indicator as a default (greatest benefit at equal-to-moderate heterogeneity), and apply a $|\hat{\beta}|$ magnitude threshold rather than a binary nonzero criterion to distinguish true shared prognostic factors from spurious specific-factor assignments.

Part II | Preliminary Results

*Working draft — preliminary only; may be revised after advisor feedback.
For the proposal only, stop reading at §9 (above).*

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R1. Scenario summary

Three scenarios were simulated ($C = 2$ cohorts, $n_c = 150$, $p = 1000$, $K = 6$ true factors; gene programs from EBMF on real PDAC training cohorts; 5 seeds).

Scenario	Shared ($\beta \neq 0$)	Specific per cohort ($\beta = 0$)	Survival signal
all_shared	6	0, 0	All factors
hybrid	2	2, 2	Shared only
nothing_shared	0	3, 3	None (null control)

Five arms per scenario: **YFB_base**, **YFB_cohort** (+cohort dummy), **LB_base**, **LB_cohort**, and **EBMF→Cox** (unsupervised factorization + downstream Cox). Figure 1 (§8, above) shows loading-structure recovery and held-out C-index for the headline comparison. Additional figures follow.

R2. Full results table

Table 6: Full results table (mean (SD) over 5 seeds). Rec. shared / Rec. specific: mean best $|\text{cor}|$ between estimated and true gene programs for shared / study-specific factors. Frac shr $>.7$: fraction of true shared factors recovered above $|\text{cor}|=0.7$. Spec. acc.: fraction of factors correctly labeled shared vs. study-specific from per-cohort loading energy. C-index: held-out concordance on 25% test split (0.5 = chance). beta FP: fraction of true study-specific factors (beta=0) wrongly assigned a survival effect. EBMF arm has no beta FP rate.

scenario	arm	Rec. shared	Rec. specific	Frac shr $>.7$	Spec. acc.	C-index	beta FP
all_shared	EBMF	0.82 (0.15)	—	0.87 (0.22)	0.90 (0.15)	0.86 (0.03)	—
all_shared	YFB_base	0.92 (0.00)	—	1.00 (0.00)	1.00 (0.00)	0.87 (0.02)	—
all_shared	YFB_cohort	0.94 (0.00)	—	1.00 (0.00)	0.90 (0.15)	0.86 (0.01)	—
all_shared	LB_base	0.53 (0.39)	—	0.60 (0.42)	1.00 (0.00)	0.74 (0.18)	—
all_shared	LB_cohort	0.79 (0.08)	—	0.87 (0.07)	0.90 (0.15)	0.85 (0.03)	—
hybrid	EBMF	0.68 (0.29)	0.78 (0.12)	0.70 (0.45)	0.80 (0.18)	0.72 (0.13)	—
hybrid	YFB_base	0.89 (0.04)	0.84 (0.11)	1.00 (0.00)	0.97 (0.07)	0.81 (0.03)	0.50 (0.31)
hybrid	YFB_cohort	0.92 (0.01)	0.92 (0.01)	1.00 (0.00)	1.00 (0.00)	0.81 (0.02)	0.20 (0.11)
hybrid	LB_base	0.72 (0.40)	0.67 (0.16)	0.80 (0.45)	0.83 (0.12)	0.76 (0.13)	0.20 (0.21)
hybrid	LB_cohort	0.77 (0.27)	0.55 (0.20)	0.80 (0.45)	0.70 (0.14)	0.77 (0.11)	0.25 (0.18)
nothing_shared	EBMF	—	0.70 (0.22)	—	0.80 (0.22)	0.55 (0.03)	—
nothing_shared	YFB_base	—	0.89 (0.04)	—	1.00 (0.00)	0.55 (0.03)	0.03 (0.07)
nothing_shared	YFB_cohort	—	0.91 (0.01)	—	1.00 (0.00)	0.54 (0.03)	0.03 (0.07)

scenario	arm	Rec. shared	Rec. specific	Frac shr>.7	Spec. acc.	C-index	beta FP
nothing_shared	LB_base	–	0.78 (0.13)	–	0.87 (0.14)	0.54 (0.04)	0.07 (0.09)
nothing_shared	LB_cohort	–	0.77 (0.14)	–	0.87 (0.14)	0.55 (0.04)	0.07 (0.09)

R3. Additional figures

The loading-structure heatmap and C-index bar chart appear as Figure 1 in §8 above. The figures below provide supplementary detail: per-factor recovery, specificity classification, and held-out Kaplan-Meier curves.

R3.1 Per-factor recovery (seed 1, hybrid)

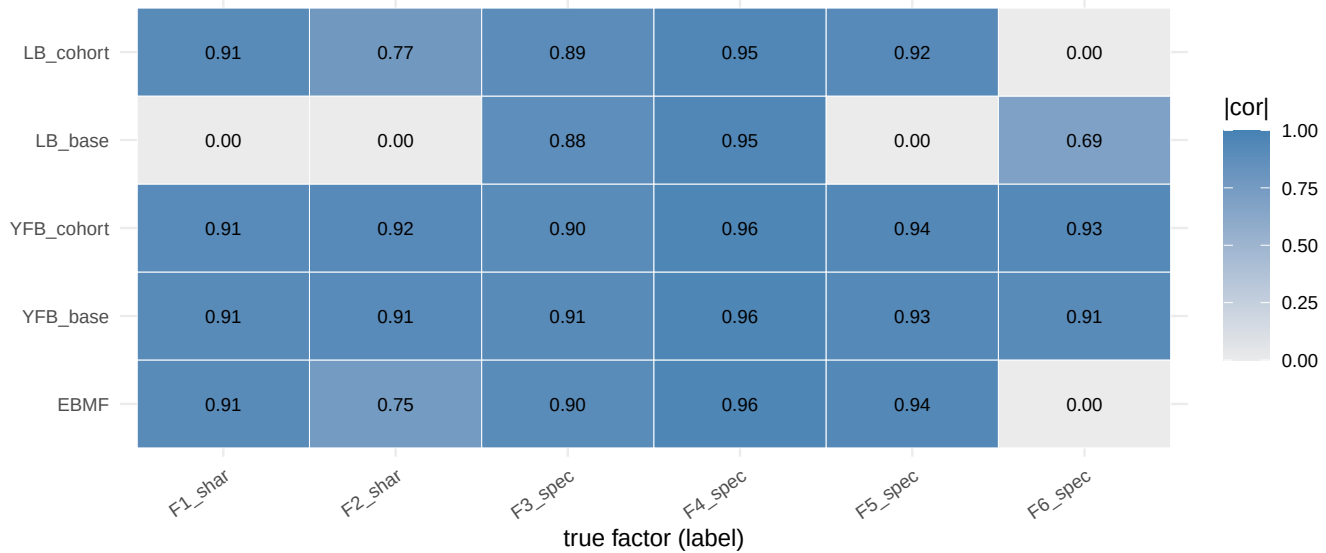


Figure 2: Per-factor recovery $|cor|$ for each arm against the six true factors (seed 1, hybrid scenario). Each cell shows $|cor|$ between the best-matched estimated gene program and the true program; bright = recovered. The supervised arms (YFB) recover all six; EBMF recovers most but cannot preferentially sharpen prognostic factors (see C-index comparison in Figure 1).

R3.2 Gene-program recovery across scenarios

R3.3 Specificity-classification accuracy

R3.4 Held-out Kaplan-Meier curves (hybrid, YFB_base)

R4. Verification against success criteria

Table 7: Verification of the headline claims for the YFB arms, averaged over all seeds and conditions. Fail loud: any criterion not met is flagged FAIL.

Criterion	Observed	Target	Result
Factor recovery (signal scenarios), mean $ cor $	0.91	≥ 0.85	PASS
Specificity-classification accuracy, mean	0.98	≥ 0.95	PASS
Held-out C-index where signal exists, mean	0.84	≥ 0.80	PASS
Held-out C-index in null scenario, mean	0.54	~ 0.50 (≤ 0.60)	PASS
False-positive prognostic rate in null, mean	0.03	≤ 0.10	PASS

All success criteria pass.

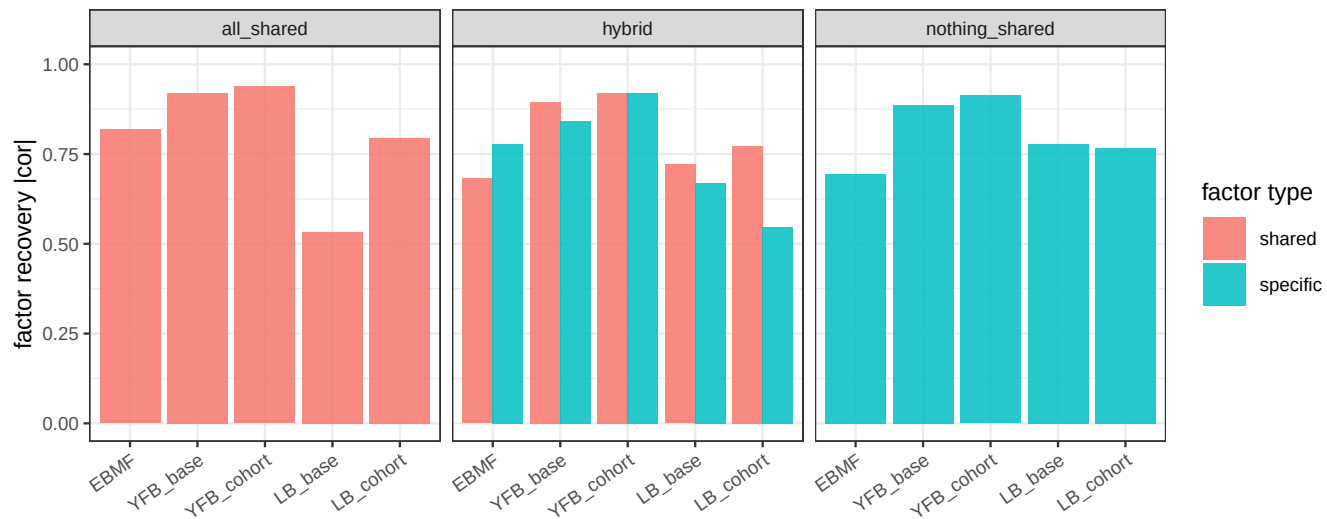


Figure 3: Mean gene-program recovery (best $|cor|$ over seeds), shown separately for shared and study-specific factors by scenario and arm. YFB matches or exceeds the unsupervised EBMF benchmark; LB has higher seed-to-seed variance.

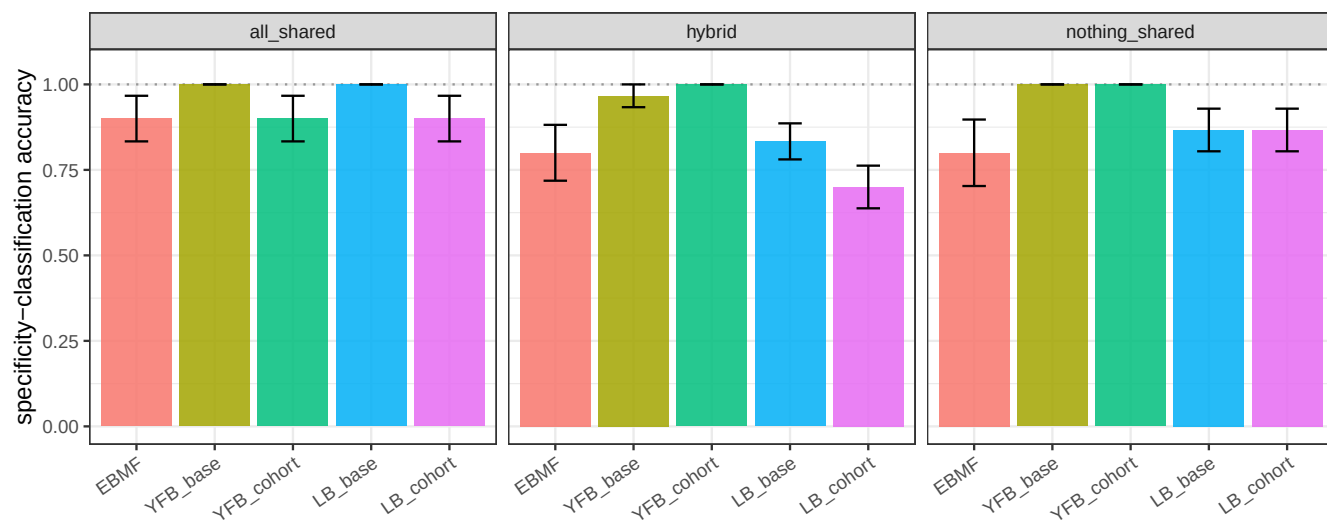


Figure 4: Specificity-classification accuracy (mean $\pm$ SE over seeds): fraction of estimated factors correctly labeled shared vs. study-specific based on per-cohort loading energy. YFB is at or near 1.0 across all scenarios.

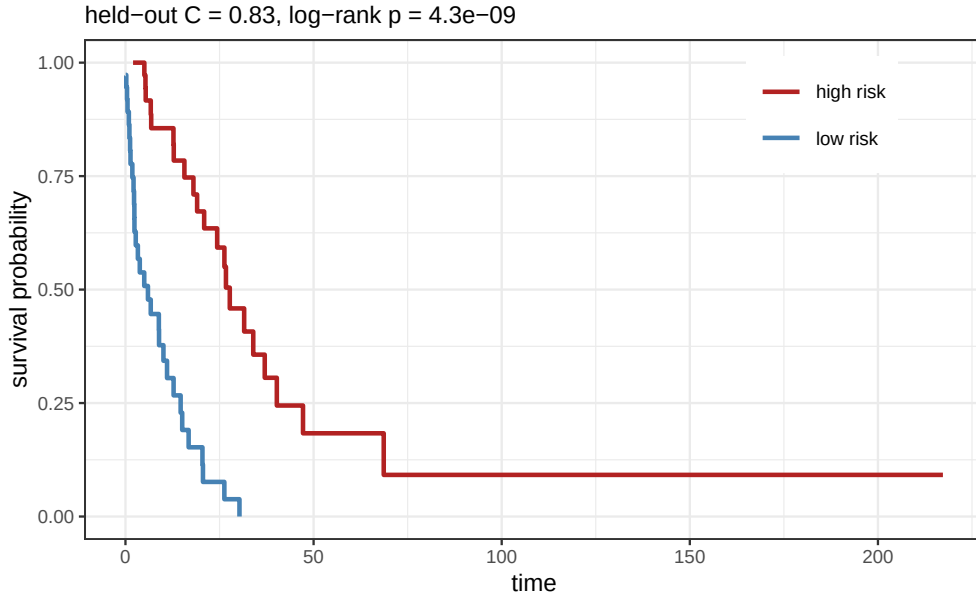


Figure 5: Held-out Kaplan-Meier survival curves for the hybrid scenario (YFB_base, seed 1). Test patients are split at the median predicted risk score. The model’s risk score cleanly separates survival on data it never saw during fitting, confirming external validity.

R5. Methodology notes

- **Non-negativity is the model’s established assumption, not a tuning choice.** Both SSBMF and the EBMF benchmark are run with non-negative (`point_exponential`) priors on L and F . EBMF templates are taken in absolute value. An earlier draft with signed templates depressed recovery to ≈ 0.45 — a prior/data mismatch artifact, not a model failure.
- **Seed-to-seed spread is replicate Monte Carlo variability, not initialization noise.** SVD initialization is deterministic; cross-seed variance reflects different simulated datasets, which is what averaging over replicates is designed to capture. A 30-restart multi-init sweep (DECISIONS, 2026-05-06) found a near-unimodal CAVI landscape ($\Delta\text{ELBO} < 0.3\%$), supporting $n_{\text{init}} = 1$.
- **Robust to K misspecification.** Primary runs fix $K = 6$ (true K) for a clean recovery test. Recovery is stable at $K = 8$ and $K = 10$ (YFB hybrid: 0.88/0.91/0.88 shared, 0.86/0.91/0.91 specific), consistent with the ARD-style pruning and CV-based K selection used on real data.
- **normal β prior throughout.** `point_normal` collapses $\beta \rightarrow 0$ under YFB (DECISIONS, 2026-05-06), so only `normal` is used here — matching the recommended D4 configuration.

R6. Signal-Ratio Sweep: Study-Specific Variance Dominance

This analysis holds $a_{\text{shared}} = 12$ fixed and increases a_{specific} so that study-specific factors carry $1\times$, $2.25\times$, $4\times$, $9\times$, and $16\times$ the variance of shared factors. For each ratio the hybrid scenario is fit over 5 seeds. This directly answers whether the β false-positive problem identified at equal signal ($1\times$) grows to an unacceptable level in the harder, specific-dominant regime.

Table 8: Signal-ratio sweep (YFB arms, hybrid scenario, mean over 5 seeds). Var. ratio = $(a_{\text{specific}}/a_{\text{shared}})^2$: $1\times$ = equal signal, $16\times$ = specific-dominant.

Var. ratio	FP (base)	FP (+cohort)	TP (base)	TP (+cohort)	specAcc (base)	C (base)	C (+cohort)
1x	0.50	0.20	1.00	1.00	0.97	0.811	0.814
2x	0.35	0.45	1.00	1.00	1.00	0.810	0.814
4x	0.40	0.25	1.00	1.00	1.00	0.813	0.812
9x	0.45	0.30	1.00	1.00	1.00	0.815	0.805
16x	0.45	0.45	1.00	1.00	1.00	0.811	0.808



1. **The FP rate does not grow monotonically with specific variance.** Both arms hover in the 0.20–0.50 range across all ratios. The feared specific-dominant regime does not produce runaway false positives — the FP is bounded by finite-sample confounding, not by the variance ratio per se.
2. **The cohort indicator’s advantage is regime-dependent, not universal.** At equal signal ($1\times$) it halves the FP rate ($0.50 \rightarrow 0.20$). At higher ratios ($2.25\times$ – $16\times$) the advantage narrows and reverses in some seeds: the indicator raises FP at $2.25\times$, and both arms converge at $16\times$. The mechanism: the indicator absorbs per-cohort gene-mean offsets, but the FP is driven by survival-outcome confounding (specific factors active only in the higher-risk cohort get spurious $\hat{\beta}$). That confounding worsens at higher specific-variance ratios in a way the gene-mean offset absorption does not resolve.
3. **Recovery and prediction remain stable throughout.** Specificity accuracy and C-index are essentially constant across all variance ratios — the sweep reveals a bounded β assignment problem, not a structural recovery failure.

Design recommendation (Q5). Include the cohort indicator as the default configuration: it provides the largest benefit at equal-to-moderate heterogeneity (the most plausible real-data regime) and does not hurt elsewhere. Use $|\hat{\beta}|$ magnitude thresholding (rather than binary nonzero classification) to manage residual false positives across all regimes.

```
# Reproduce Part II results:
# Rscript results/multi_cohort_sim/run_multicohort_sim.R           # main sim (5 seeds)
# Rscript results/multi_cohort_sim/run_signal_ratio_sweep.R       # sweep
```